Pre-Analysis Plan - Team RT

#ManyDaughters: Multi-Analyst Project Exploring the Effect of Daughters on Preferences and Behaviors

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Introduction

This pre-analysis plan (PAP) is prepared as part of the *ManyDaughters* project, following the guidelines for research teams assigned to the no-data treatment group. In accordance with the protocol, this document has been written without access to the SOEP dataset and outlines the planned empirical strategy, variable construction, and hypotheses to be tested.

The goal of this project is to test four causal hypotheses about how having daughters influences parents' attitudes toward gender equality, political preferences, and household behavior. Previous studies suggest that parenting daughters can shape parents' beliefs and actions related to gender norms and equality (e.g., Washington (2008); Shafer and Malhotra (2011); Sharrow et al. (2018); Borrell-Porta et al. (2019); Greenlee et al. (2020)). We aim to replicate and extend these findings using data from the German Socio-Economic Panel (SOEP). Our analysis is constrained by data availability to a subset of SOEP datasets, namely: `pl`, `hl`, `biol`, `biobirth`, `bioparen`, `pgen`, `hgen`, and `ppathl`, covering the years 2000 to 2020.

General identification strategy: Following Greenlee et al. (2020), we define the treatment as parenting a first-born daughter, restricting the sample to respondents whose first birth was a singleton (i.e., excluding multiple births such as twins). This approach allows us to estimate the effect of having a daughter versus a son on attitudes toward gender equality, net of the broader effect of becoming a parent. We focus on the sex of the first child to mitigate concerns related to fertility-stopping behavior, as the sex of the first child is arguably the most exogenous measure. This choice strengthens the causal interpretation of our estimates. Parents may make subsequent fertility decisions based on the sex composition of their existing children, introducing potential endogeneity when using variables such as "having at least one daughter." Indeed, evidence from Western

countries suggests a preference for mixed-sex siblings (Lundberg, 2005). Recent research also highlights the unique socializing effect of fathering a daughter, particularly as a first child. For example, Sharrow et al. (2018) finds that having a first daughter is associated with more liberal gender attitudes among fathers, an effect not observed when considering the general experience of having daughters or the proportion of daughters in the family.

Our sample will be restricted to adults (aged 18 and older) who are biological parents and were interviewed in waves between 2000 and 2020, retaining only observations from years in which they had at least one child. All models include year fixed effects and control for the number of children. Standard errors will be clustered at the individual or household level. Tables throughout this document summarize the variables used and how they will be constructed once access to the data is granted.

Hypothesis 1: Gender Equality Attitudes

Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality,

$$\mathbb{E}[X \mid \text{has daughter}] > \mathbb{E}[X \mid \text{no daughter}]$$

Sample selection

For Hypothesis 1, we restrict the sample to respondents observed in the 2018 wave, which includes a survey question about respondents' attitudes toward gender equality. We include respondent who are biological parents at the time of the interview ($\text{biolparent}_{i,2018} = 1$, as described in Table 1).

Empirical Strategy

We estimate the following OLS specification:

$$X_i = \alpha + \beta_1 \cdot \text{hasdaughter}_i + \sum_{k=2}^{4} \beta_{2k} \cdot \text{child}_{k,i} + \beta_3 \cdot \text{controls}_i + \varepsilon_i \tag{1}$$

Where:

- X_i: Gender equality attitudes score, where higher values indicate more egalitarian views.
- hasdaughter_i: Treatment variable; an indicator equal to 1 if the respondent's first child is a daughter, and 0 if the first child is a son (by the time of the interview).

- $\sum_{k=2}^{4} \text{child}_{k,i}$ denotes a set of three dummy variables indicating whether the respondent has two, three, or four or more children. The reference category is having exactly one child, which is omitted from the equation.

- controls_i: Vector of covariates controlling for demographic and socioeconomic characteristics, including three dummies for the age of the first children, age, age^2, male, two education dummies, five marital status dummies, an employment status dummy, the logarithm of household income, and a dummy for being born in Germany.

- ε_i: Heteroskedasticity-robust standard errors.

To operationalize the outcome variable X_i, we use responses from the gender attitude question available in the SOEP 2018 (`plh0308_v2`): *"It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally."* (from 1=Do not agree at all to 7=Completely agree). This question captures attitudes toward gender equality in work and household roles. Table 2 presents the exact survey item used to define X_{it}. The model is estimated using Ordinary Least Squares (OLS) with heteroskedasticity-robust standard errors.

Our key explanatory variable is a treatment indicator based on the gender of the respondent's first biological child. Specifically, we construct the variable hasdaughter_i, a binary indicator equal to 1 if individual i's first child is female. This measure is considered more plausibly exogenous than simply having at least one daughter, as the gender of the first child is less likely to be influenced by parental preferences or fertility decisions. The construction of this variable is described in Table 1.

Our specification includes a comprehensive set of control variables. This includes a set of dummy variables for the number of children, indicating whether the respondent has two, three, or four or more children (with one child as the reference category). It also includes a vector of demographic and socioeconomic covariates: three dummies for the age of the first children (0–4, 5–9, 10–14, older than 14 as reference), age, age^2, a dummy equal to 1 if the respondent is male (0 if female), two education dummies (upper secondary and tertiary education), five marital status dummies, an employment status dummy, the logarithm of household income, a dummy indicating whether the respondent was born in Germany, and a disability indicator. These controls aim to account for potential confounding factors affecting both family composition and the outcome variables. Table 3 describes the construction of all control variables.[1] The error term ε_i captures unobserved factors affecting the outcome for individual i.

[1]For Hypotheses 1, 2, and 4, we closely follow the control variables used in Borrell-Porta et al. (2019), with one exception: regional controls. Including a distinction between East and West Germany could be particularly relevant, as studies such as Jessen (2022) document persistent regional differences in gender norms and behaviors. We considered including an East/West Germany indicator based on the respondent's region of residence for all our hypothesis using the variable (`ppathl:sampreg`). However, this variable appears to be available only up to 2017. If this turns out to be a misunderstanding, and consistent regional information is available beyond 2017, we will include it accordingly.

Interpretation of the results A positive coefficient on the treatment variable, $\beta_1 > 0$, provides support for the hypothesis that having daughters increases support for gender equality. We will interpret the estimate as evidence supporting the hypothesis if β_1 is positive and statistically significant at the 5% level, based on the two-sided t-statistic from the regression output. We will report the coefficient, its t-statistic, the corresponding p-value, and the 95% confidence interval for reference.

Hypothesis 2: Political Preferences

Hypothesis: Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences,

$$\mathbb{E}[Y \mid \text{has daughter}] > \mathbb{E}[Y \mid \text{no daughter}]$$

Sample selection

For Hypothesis 2, we restrict the sample to respondents observed in waves 2005, 2009, 2014, and 2019, when the dependent variable is available. We include only biological parents ($\text{biolparent}_{it} = 1$, as described in Table 1).

Empirical Strategy

We estimate the following pooled OLS baseline specification:

$$Y_{it} = \alpha + \beta_1 \cdot \text{hasdaughter}_{it} + \sum_{k=2}^{4} \beta_{2k} \cdot \text{child}_{k,it} + \beta_3 \cdot \text{controls}_{it} + \delta_t + \varepsilon_{it} \qquad (2)$$

Where:

- Y_{it}: Political preferences score, where higher values indicate more left-leaning views.
- hasdaughter_{it}: Treatment variable; an indicator equal to 1 if the respondent's first child is a daughter, and 0 if the first child is a son, at time t.
- $\sum_{k=2}^{4} \text{child}_{k,it}$ denotes a set of three dummy variables indicating whether the respondent has two, three, or four or more children. The reference category is having exactly one child, which is omitted from the equation.
- controls_{it}: Vector of covariates controlling for demographic and socioeconomic characteristics, including three dummies for the age of the first children, age, age^2, male, two education dummies, five marital status dummies, an employment status dummy, the logarithm of household income, and a dummy for being born in Germany.

- δ_t: Year fixed effects.
- ε_{it}: Error term, clustered at the individual level.

To operationalize the outcome variable Y_{it}, we use respondents' self-placement on a political left–right scale, based on the SOEP variable `plh0004`. This question is asked in selected waves (2005, 2009, 2014, 2019) and invites individuals to place themselves on an 11-point scale, ranging from 0 (left) to 10 (right). To facilitate interpretation, we reverse-code the responses as $11 - $ `plh0004`, so that higher values correspond to more left-leaning political preferences. Table 2 details the coding strategy and construction of the political preferences variable.

The key explanatory variable and control set used in this analysis are identical to those employed in Hypothesis 1. We estimate a pooled Ordinary Least Squares (OLS), with year (wave) fixed effects δ_t and standard errors ε_{it} clustered at the individual level for all models.

Interpretation of the results A positive coefficient on the treatment variable, $\beta_1 > 0$, provides support for the hypothesis that having daughters leads to more left-leaning political preferences. We will interpret the estimate as evidence supporting the hypothesis if β_1 is positive and statistically significant at the 5% level, based on the two-sided t-statistic from the regression output. We will report the coefficient, its t-statistic, the corresponding p-value, and the 95% confidence interval for reference.

Hypothesis 3: Gender Equality in the Household

Hypothesis: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household,

$$\mathbb{E}[Z \mid \text{has daughter}] > \mathbb{E}[Z \mid \text{no daughter}]$$

Sample selection : Our analysis is based on a restricted sample drawn from the German Socio-Economic Panel (SOEP) covering the years 2000 to 2020. As we aim to investigate gender equality within couples, we restrict the sample to couples who are biological parents. We follow Jessen (2022) and include both married and non-married cohabiting couples (cohab_{it}=1, as described in Table 1) and both at working age (18-65 yo).

We further restrict the sample to couples who share the same first biological child, ensuring that the transition to parenthood represents a joint experience. This restriction excludes blended families and mitigates concerns about endogenous partner matching, where individuals with more gender-equal attitudes may self-select into relationships with

certain types of partners or preexisting children. By doing so, we enhance the consistency of the treatment definition and strengthen the causal interpretation of our estimates. Specifically, we proceed as follows:

- Use `kidpnr01` to identify the personal identifiers (`pid`) of the mother and father of each respondent's first child.
- Retain only cohabiting couples who are jointly listed as biological parents of the same first child.

To ensure accurate computation of gender-based outcomes (e.g., division of housework), we further exclude respondents in same-sex partnerships. As our metrics will refer to time-use doing domestic work, we exclude couples with individuals with at least one registered disability using the SOEP disability indicator. The variables used to select the sample are described in Table 1.

Empirical Strategy

$$Z_{ht} = \alpha + \beta_1 \cdot \text{hasdaughter}_{ht} + \beta_2 \cdot \text{controls}_{ht} + \delta_t + \varepsilon_{ht} \tag{3}$$

Where:

- Z_{ht}: Proxy for gender equality in household h at time t.
- hasdaughter_{ht}: Treatment variable; an indicator equal to 1 if the couple's first child in the household is a daughter, and 0 if the first child is a son, at time t.
- controls_{ht}: A vector of covariates controls for demographic and socioeconomic characteristics, including the mother's and father's age and age squared; indicators for whether both partners have the same level of education and the highest level of education attained within the couple; a dummy for being a married couple; an indicator for both partners being employed; a dummy for whether the mother is the primary earner; the logarithm of household income; dummies for both partners being born in Germany and for both not being born in Germany; and the number of children living in the household, categorized into three age brackets: 0–4, 5–9, and 10–14 years.
- δ_t: Year fixed effects.
- ε_{ht}: Error term, clustered at the household level.

To operationalize the outcome variable Z_{it}, we consider the unpaid domestic work distribution within the couple. As suggested by Goldin (2014), gender wage disparities persist not merely because of differences in paid-labor participation but crucially because of unpaid domestic labor (particularly housework and childcare), which remains unequally shared.

We will look at contributions to housework and childcare. Specifically, the questionnaire asks how many hours respondents normally spend on these tasks on weekdays. To test our hypothesis, we built an equality index by computing the ratio of the smaller to the larger value of domestic work hours between the respondent and their cohabiting partner. Formally, this is defined as:

$$Z_{it} = \frac{\min(\text{DW}_{it},\ \text{DW}_{it}^{\text{partner}})}{\max(\text{DW}_{it},\ \text{DW}_{it}^{\text{partner}})}$$

This ratio ranges from 0 (complete inequality) to 1 (perfect equality), with higher values indicating a more balanced distribution of household responsibilities. We exclude cases where either the respondent or their partner reports a negative or invalid value for housework hours. Table 2 describes the construction of the variable. Our main analysis uses this ratio as a continuous dependent variable, and the model is estimated using Ordinary Least Squares (OLS). Control variables are described in Table 4.[2]

Interpretation of the results: A positive coefficient on the treatment variable, $\beta_1 > 0$, would support the hypothesis that having daughters is associated with greater gender equality, as measured by a more equitable division of unpaid domestic work within the household. We will interpret the estimate as evidence supporting the hypothesis if β_1 is positive and statistically significant at the 5% level, based on the two-sided t-statistic from the regression output. We will report the coefficient, its t-statistic, the corresponding p-value, and the 95% confidence interval for reference.

Hypothesis 4: Gender Differences in the Impact of Daughters on Equality Attitudes

Hypothesis: Having daughters leads to a larger increase in the support for gender equality among men than among women. That is, denoting X as a measure of support for gender equality:

$$(\mathbb{E}[X \mid \text{has daughter, male}] - \mathbb{E}[X \mid \text{no daughter, male}]) >$$
$$(\mathbb{E}[X \mid \text{has daughter, female}] - \mathbb{E}[X \mid \text{no daughter, female}])$$

Sample selection

The sample selection in this analysis is identical to those employed in Hypothesis 1.

[2]We follow standard practice in the literature in selecting control variables, as in Jessen (2022) and Siminski and Yetsenga (2022).

Empirical Strategy

We test this hypothesis by adding an interaction term between the treatment variable and a male dummy using the following OLS specification:

$$X_i = \alpha + \beta_1 \cdot \text{hasdaughter}_i + \beta_2 \cdot \text{male}_i + \beta_3 \cdot (\text{hasdaughter}_i \times \text{male}_i) + \sum_{k=2}^{4} \beta_{4k} \cdot \text{child}_{k,i} + \beta_5 \cdot \text{controls}_i + \varepsilon_i \quad (4)$$

Where:

- X_i: Gender equality attitudes score, where higher values indicate more egalitarian views.
- hasdaughter_i: Treatment variable; an indicator equal to 1 if the couple's first child in the household is a daughter, and 0 if the first child is a son, at time t.
- $\sum_{k=2}^{4} \text{child}_{k,i}$ denotes a set of three dummy variables indicating whether the respondent has two, three, or four or more children. The reference category is having exactly one child, which is omitted from the equation.
- controls_i: Vector of covariates controlling for demographic and socioeconomic characteristics, including three dummies for the age of the first children, age, age^2, two education dummies, five marital status dummies, an employment status dummy, the logarithm of household income, and a dummy for being born in Germany.
- ε_i: Heteroskedasticity-robust standard errors.

Interpretation of the results: The coefficient of interest is β_3. A positive and significant β_3 would support Hypothesis 4, indicating that the effect of having daughters on support for gender equality is stronger among fathers than among mothers. We will interpret the estimate as evidence supporting the hypothesis if β_3 is positive and statistically significant at the 5% level, based on the two-sided t-statistic from the regression output. We will report the coefficient, its t-statistic, the corresponding p-value, and the 95% confidence interval for reference.

Tables

Table 1: Sample selection and treatment variables

Name	Description	SOEP code	Score type	Years
biolparent$_{it}$	Biological parent. Dummy variable equal to 1 if the respondent has at least one child at time t: `kidgeb01` $\leq t$.	`biobirth`: `kidgeb01`.	Computed	2000–2020
hasdaughter$_{it}$	Dummy variable equal to 1 if the respondent's first biological child is a daughter at time t, based on `kidsex01` = 2 and `kidgeb01` $\leq t$	`biobirth`: `kidsex01`, `kidgeb01`.	Computed	2000–2020
numchildren$_{it}$	Number of biological children. Count of the respondent's biological children at time t. Computed by counting valid values (1 = male, 2 = female, 3 = diverse) in `kidsex01`—`kidsex19` and assuming `kidgeb0*` $\leq t$. Values from -1 to -7 are excluded.	`biobirth`: `kidsex0-kidsex19`.	Computed	2000–2020
cohab$_{it}$	Cohabiting with partner. Dummy variable equal to 1 if the respondent lives with a spouse or cohabiting partner in the same household at time t, identified by a valid partner ID in `pgpartnr` ($\geq$ 1). It is set to 0 if `pgpartnr` = -2 ("does not apply," i.e., no spouse or cohabiting partner). Cases with other negative values are excluded.	`pgen`: `pgpartnr`	Computed	2000–2020
samesex$_{it}$	Dummy =1 if respondent and partner are of same sexes. Constructed by linking `pla0009_h` for respondent and `pla0009_h` for partner (via pgpartnr).	`pgen`: `pla0009_h`, `pgen`: `pgpartnr`	Computed	2000–2020
disability$_{it}$	Dummy equal to 1 if the respondent is legally certified as disabled or reduced in employment capacity (`ple0040` = 1), O otherwise. Negative codes are excluded.	`pl`: `ple0040`	Computed	2000–2020

Note: The SOEP Companion recommends two variables for the identification of partners: `parid` from `ppathl` and `pgpartnr` from `pgen`. Since this document was prepared without access to the full data, we refer to `pgpartnr` in our construction. However, `parid` may also be used if the two variables are not strictly identical.

Table 2: Dependent variables

Name	Description	SOEP code	Score type	Years
Gender equality attitudes score (X_{it})				
genderattitudes$_i$	Gender division of work: *"It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally."*; 1 = strongly disagree, 7 = strongly agree (7-point Likert scale).	`pl: plh0308_v2`	Original	2018
Political preferences (Y_{it})				
lrscale$_{it}$	Political self-placement on a left–right scale. Original responses range from 0 (left) to 10 (right). Reverse-coded as 11 − `plh0004`, so that higher scores indicate more left-leaning preferences.	`pl: plh0004`	Recoded	2005, 2009, 2014, 2019
Household Gender Equality Proxy (Z_{it})				
housework$_{it}$	Respondent's reported time spent on housework per working day (in hours). Negative codes are excluded.	`pl: pli0043_v3`	Original	2000–2020
housework$_{it}^{\text{partner}}$	Partner's reported housework hours per working day. Computed by linking to the partner's response using `pgpartnr` (partner's ID). Negative codes are excluded.	`pl: pli0043_v3`, `pgen: pgpartnr`	Original	2000–2020
childcare$_{it}$	Respondent's reported time spent on child care per working day (in hours). Negative codes are excluded.	`pl: pli0044_v3`	Original	2000–2020
childcare$_{it}^{\text{partner}}$	Partner's reported time spent on child care per working day. Computed by linking to the partner's response using `pgpartnr` (partner's ID). Negative codes are excluded.	`pl: pli0044_v3`, `pgen: pgpartnr`	Original	2000–2020
DW$_{it}$	Unpaid domestic work, computed by: DW$_{it}$ = housework$_{it}$ + childcare$_{it}$	—	Computed	2000–2020
DW$_{it}^{\text{partner}}$	Partner's unpaid domestic work, computed by: DW$_{it}^{\text{partner}}$ = housework$_{it}^{\text{partner}}$ + childcare$_{it}^{\text{partner}}$	—	Computed	2000–2020

Note: Variables `pl: pli0043_v3` (`pl: pli0043_h`) and `pl: pli0044_v3` (`pl: pli0044_h`) have identical or very similar definitions. Once the data are available, we will assess equivalence and may use the versions in parentheses when appropriate; that is, if they increase the number of valid observations or better cover the study period.

Table 3: Control variables - individual level

Name	Description	SOEP code	Score type	Years
child2$_{it}$	Dummy for two children. Equals 1 if numchildren$_{it}$ = 2; 0 otherwise. The variable numchildren$_{it}$ is defined on Table 1.	—	Computed	2000–2020
child3$_{it}$	Dummy for three children. Equals 1 if numchildren$_{it}$ = 3; 0 otherwise.	—	Computed	2000–2020
child4plus$_{it}$	Dummy for four or more children. Equals 1 if numchildren$_{it} \geq 4$; 0 otherwise.	—	Computed	2000–2020
first0-4$_{it}$	Dummy equal to 1 if the respondent's first child is aged 0–4 at time t; 0 otherwise. Computed as survey year − `kidgeb01` $\in [0, 4]$.	`biobirth: kidgeb01, syear`	Computed	2000–2020
first5-9$_{it}$	Dummy equal to 1 if the respondent's first child is aged 5–9 at time t; 0 otherwise. Computed as `syear` − `kidgeb01` $\in [5, 9]$.	`biobirth: kidgeb01, syear`	Computed	2000–2020
first10-14$_{it}$	Dummy equal to 1 if the respondent's first child is aged 10–14 at time t; 0 otherwise. Computed as survey year − `kidgeb01` $\in [10, 14]$.	`biobirth: kidgeb01, syear`	Computed	2000–2020
age$_{it}$	Respondent's age in years at time t, computed as `syear` − `gebjahr`.	`pl: gebjahr, syear`	Computed	2000–2020
age2$_{it}$	Squared age	`pl: gebjahr, syear`	Computed	2000–2020
male$_i$	Respondent's sex: dummy = 1 if male (`pla0009_h` = 1), female = 0 if `pla0009_h` = 2; negative/missing codes are excluded.	`pl: pla0009_h`	Computed	2000–2020
Education dummies	Set of two education dummies based on `pgcasmin`. The dummy *edu_high*$_{it}$ equals 1 for upper secondary education (CASMIN values 6–7); *edu_tertiary*$_{it}$ equals 1 for tertiary education (CASMIN values 8–9). The reference category includes respondents with lower secondary education or less (CASMIN values 0–5).	`pgen: pgcasmin`	Recoded	2000–2020
Marital status dummies	Set of five dummies for marital status constructed from `pgfamstd`: *married*$_{it}$ (1), *married_sep*$_{it}$ (2), *divorced*$_{it}$ (4), *widowed*$_{it}$ (5), and *partner_abroad*$_{it}$ (6); the reference category is single (3). Respondents coded as 7 or 8 are excluded.	`pgen: pgfamstd`	Recoded	2000–2020
employed$_{it}$	Employment status dummy variable constructed from `pgemplst`. *employed*$_{it}$ = 1 if the respondent is employed full-time or part-time (codes 1 or 2); 0 if not employed (codes 3–7).	`pgen: pgemplst`	Recoded	2000–2020
ln_hhinc$_{it}$	Log of total net household monthly income: ln(`HINC$$` + 1). Based on imputed SOEP variable in `hgen`. Negative/missing codes excluded.	`hgen: HINC$$`	Computed	2000–2020
germanborn$_i$	Dummy variable equal to 1 if respondent was born in Germany or immigrated before 1950 (i.e., `germborn` = 1).	`ppathl: germborn`	Computed	2000–2020

Note: The variables listed were selected prior to accessing the data. Once the empirical analysis begins, we may replace them with closely related alternatives in parentheses, when appropriate; to increase coverage or improve data availability. These include: `pl: pla0009` (`pla0009_v2`, `pla0009_v3`, `ppathl: sex`); `pgen: pgcasmin` (`pgpsbil`, `pgpbbil02`); `pgen: pgfamstd` (`ppathl: partner`); `hgen: HINC$$` (`pglabnet` + `pglabnet` [partner]). The variable `ppathl: germborn` may be excluded from the final specification if its inclusion leads to substantial sample size reductions. We considered including an East/West Germany indicator based on the respondent's region of residence for all our hypothesis using the variable (`ppathl:sampreg`). However, this variable appears to be available only up to 2017. If this turns out to be a misunderstanding, and consistent regional information is available beyond 2017, we will include it in the specifications for all the hypothesis.

Table 4: Control variables - household level

Name	Description	SOEP code	Score type	Years
sameeduc$_{ht}$	Dummy variable equal to 1 if the respondent and their partner belong to the same education category, based on the three-category classification defined in the two dummies (*edu_high*, *edu_tertiary*) described in Table 3. Partner education is linked via `pgpartnr`.	`pgen: pgcasmin, pgpartnr`	Computed	2000–2020
Highest education dummies	Two dummy variables for the highest education level in the household. Based on the education dummies defined in Table 3, we compare the respondent's and their partner's education levels (identified via `pgpartnr`). *edu_tertiary*$_{ht}$ equals 1 if either partner has tertiary education; *edu_high*$_{ht}$ equals 1 if neither has tertiary education but at least one has upper secondary education. The reference category includes households where both partners have lower secondary education or less.	`pgen: pgcasmin, pgpartnr`	Computed	2000–2020
married$_{ht}$	Dummy variable equal to 1 if both partners are coded as *married*$_{it}$ in the marital status dummies described in Table 3. The variable is constructed using respondent and partner information via `pgpartnr`.	`pgen: pgfamstd, pgpartnr`	Computed	2000–2020
Work combination dummies	Set of three dummy variables capturing employment configurations in the household, based on the individual employment dummy *employed*$_{it}$ defined in Table 3. Using this dummy for both the respondent and their partner (identified via `pgpartnr`), we define: *both_working*$_{ht} = 1$ if both are employed; *male_breadwinner*$_{ht} = 1$ if only the male partner is employed; *female_breadwinner*$_{ht} = 1$ if only the female partner is employed. The reference category includes couples in which neither partner is employed.	`pgen: pgemplst, pgpartnr`	Recoded	2000–2020
Born in Germany dummies	Set of two dummy variables capturing the couple's birthplace configuration. *both_born_in_germany*$_{ht} = 1$ if both the respondent and their partner (identified via `pgpartnr`) were born in Germany; *none_born_in_germany*$_{ht} = 1$ if neither was born in Germany. The reference category includes couples in which only one partner was born in Germany. Based on the individual dummy *born_in_germany*$_{it}$ defined in Table 3.	`ppathl: germborn, pgpartnr`	Recoded	2000–2020
Children aged 0–4	Number of children in the household aged 0–4 at time t, computed using all available child birth year variables `kidgeb01`–`kidgeb19` and `syear`. Each child is counted if `syear`$-$`kidgebXX` $\in [0, 4]$.	`biobirth: kidgeb01-kidgeb19, syear`	Computed	2000–2020
Children aged 5–9	Number of children in the household aged 5–9 at time t, based on `kidgeb01`–`kidgeb19` and `syear`. Each child is counted if `syear`$-$`kidgebXX` $\in [5, 9]$.	`biobirth: kidgeb01-kidgeb19, syear`	Computed	2000–2020
Children aged 10–14	Number of children in the household aged 10–14 at time t, based on `kidgeb01`–`kidgeb19` and `syear`. Each child is counted if `syear` $-$ `kidgebXX` $\in [10, 14]$.	`biobirth: kidgeb01-kidgeb19, syear`	Computed	2000–2020

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